

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk:

Product Description:

Cat No. :

Lead zirconium titanium oxide, polymeric precursor
Lead zirconium titanium oxide, polymeric precursor

39758

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use

Laboratory chemicals.

Uses advised against

No Information available

**Company**

 Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
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 Selangor Darul Ehsan, Malaysia.  
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Tel: +03-5525 7888

 CHEMTREC Malaysia **1-800-815-308** (Malay)

 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                      |                      |
|------------------------------------------------------|----------------------|
| Aspiration Toxicity                                  | Category 1 (H304)    |
| Acute oral toxicity                                  | Category 4 (H302)    |
| Acute Inhalation Toxicity - Vapors                   | Category 4 (H332)    |
| Skin Corrosion/Irritation                            | Category 2 (H315)    |
| Serious Eye Damage/Eye Irritation                    | Category 2 (H319)    |
| Reproductive Toxicity                                | Category 1A (H360Df) |
| Specific target organ toxicity - (repeated exposure) | Category 2 (H373)    |
| Acute aquatic toxicity                               | Category 1 (H400)    |
| Chronic aquatic toxicity                             | Category 1 (H410)    |

**Label Elements**


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## Signal Word

## Danger

### Hazard Statements

H302 + H332 - Harmful if swallowed or if inhaled  
H304 - May be fatal if swallowed and enters airways  
H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H360Df - May damage the unborn child. Suspected of damaging fertility  
H373 - May cause damage to organs through prolonged or repeated exposure  
H410 - Very toxic to aquatic life with long lasting effects

### Precautionary Statements

#### Prevention

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear eye protection/ face protection

#### Response

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P332 + P313 - If skin irritation occurs: Get medical advice/attention  
P362 + P364 - Take off contaminated clothing and wash it before reuse

#### Storage

P403 - Store in a well-ventilated place

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                                                | CAS No     | Weight % |
|----------------------------------------------------------|------------|----------|
| Lead titanium zirconium oxide (Pb(Ti,Zr)O <sub>3</sub> ) | 12626-81-2 | 80       |
| Toluene                                                  | 108-88-3   | 10       |
| Isopropyl alcohol                                        | 67-63-0    | 10       |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

If symptoms persist, call a physician.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get

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medical attention.

## Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

## Ingestion

Clean mouth with water and drink afterwards plenty of water. Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.

## Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur. Risk of serious damage to the lungs (by aspiration).

## Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

## Most important symptoms and effects, both acute and delayed

None reasonably foreseeable. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

## Indication of any immediate medical attention and special treatment needed

### Notes to Physician

Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Metal fumes and oxides.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required.

### Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

### Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

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## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component                                                | Malaysia | ACGIH TLV                                                                          | OSHA PEL                                                                                                                                                                          |
|----------------------------------------------------------|----------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lead titanium zirconium oxide (Pb(Ti,Zr)O <sub>3</sub> ) |          | TWA: 0.05 mg/m <sup>3</sup> TWA: 5 mg/m <sup>3</sup><br>STEL: 10 mg/m <sup>3</sup> | (Vacated) TWA: 5 mg/m <sup>3</sup><br>(Vacated) STEL: 10 mg/m <sup>3</sup>                                                                                                        |
| Toluene                                                  |          | TWA: 20 ppm                                                                        | (Vacated) TWA: 100 ppm<br>(Vacated) TWA: 375 mg/m <sup>3</sup><br>Ceiling: 300 ppm<br>(Vacated) STEL: 150 ppm<br>(Vacated) STEL: 560 mg/m <sup>3</sup><br>TWA: 200 ppm            |
| Isopropyl alcohol                                        |          | TWA: 200 ppm<br>STEL: 400 ppm                                                      | (Vacated) TWA: 400 ppm<br>(Vacated) TWA: 980 mg/m <sup>3</sup><br>(Vacated) STEL: 500 ppm<br>(Vacated) STEL: 1225 mg/m <sup>3</sup><br>TWA: 400 ppm<br>TWA: 980 mg/m <sup>3</sup> |

| Component                                                | European Union                                                                                                                | The United Kingdom                                                                                                                            | Germany                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lead titanium zirconium oxide (Pb(Ti,Zr)O <sub>3</sub> ) |                                                                                                                               | STEL: 10 mg/m <sup>3</sup> 15 min<br>TWA: 5 mg/m <sup>3</sup> 8 hr<br>STEL: 0.45 mg/m <sup>3</sup> 15 min<br>TWA: 0.15 mg/m <sup>3</sup> 8 hr | TWA: 0.004 mg/m <sup>3</sup> (8 Stunden).<br>MAK except lead arsenate and lead chromate<br>Höhepunkt: 0.032 mg/m <sup>3</sup>                                                                                                                                              |
| Toluene                                                  | TWA: 50 ppm (8hr)<br>TWA: 192 mg/m <sup>3</sup> (8hr)<br>STEL: 100 ppm (15min)<br>STEL: 384 mg/m <sup>3</sup> (15min)<br>Skin | STEL: 100 ppm 15 min<br>STEL: 384 mg/m <sup>3</sup> 15 min<br>TWA: 50 ppm 8 hr<br>TWA: 191 mg/m <sup>3</sup> 8 hr<br>Skin                     | TWA: 50 ppm (8 Stunden). AGW -<br>exposure factor 2<br>TWA: 190 mg/m <sup>3</sup> (8 Stunden). AGW<br>- exposure factor 2<br>TWA: 50 ppm (8 Stunden). MAK<br>TWA: 190 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 100 ppm<br>Höhepunkt: 380 mg/m <sup>3</sup><br>Haut |
| Isopropyl alcohol                                        |                                                                                                                               | STEL: 500 ppm 15 min<br>STEL: 1250 mg/m <sup>3</sup> 15 min<br>TWA: 400 ppm 8 hr<br>TWA: 999 mg/m <sup>3</sup> 8 hr                           | TWA: 200 ppm (8 Stunden). AGW -<br>exposure factor 2<br>TWA: 500 mg/m <sup>3</sup> (8 Stunden). AGW<br>- exposure factor 2<br>TWA: 200 ppm (8 Stunden). MAK<br>TWA: 500 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 400 ppm<br>Höhepunkt: 1000 mg/m <sup>3</sup>      |

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## Exposure Controls

### **Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### **Personal protective equipment**

#### **Eye Protection**

Goggles

#### **Hand Protection**

Protective gloves

#### **Skin and body protection**

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g.

sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

#### **Recommended Filter type:**

low boiling organic solvent Type AX Brown conforming to EN371 or Organic gases and vapours filter Type A Brown conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice

### **Environmental exposure controls**

Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

#### **Appearance**

Pale yellow

#### **Physical State**

Liquid Viscous liquid

#### **Odor**

No information available

#### **Odor Threshold**

No data available

#### **pH**

Not applicable

#### **Melting Point/Range**

No data available

#### **Softening Point**

No data available

#### **Boiling Point/Range**

No information available

#### **Flash Point**

4 °C / 39.2 °F

**Method -** No information available

#### **Evaporation Rate**

No data available

#### **Flammability (solid,gas)**

Not applicable

Liquid

#### **Explosion Limits**

**Lower** 1.1%

**Upper** 12.0%

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|                              |                          |             |
|------------------------------|--------------------------|-------------|
| Vapor Pressure               | No data available        |             |
| Vapor Density                | No data available        | (Air = 1.0) |
| Specific Gravity / Density   | No data available        |             |
| Bulk Density                 | Not applicable           | Liquid      |
| Water Solubility             | Hydrolyzes               |             |
| Solubility in other solvents | No information available |             |

## Partition Coefficient (n-octanol/water)

| Component         | log Pow |
|-------------------|---------|
| Toluene           | 2.73    |
| Isopropyl alcohol | 0.05    |

|                           |                          |                                             |
|---------------------------|--------------------------|---------------------------------------------|
| Autoignition Temperature  | 455 °C / 851 °F          |                                             |
| Decomposition Temperature | No data available        |                                             |
| Viscosity                 | No data available        |                                             |
| Explosive Properties      |                          | Vapors may form explosive mixtures with air |
| Oxidizing Properties      | No information available |                                             |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

|                          |                               |
|--------------------------|-------------------------------|
| Hazardous Polymerization | No information available.     |
| Hazardous Reactions      | None under normal processing. |

### Conditions to Avoid

Exposure to moist air or water. Keep away from open flames, hot surfaces and sources of ignition.

### Incompatible Materials

Strong oxidizing agents. Water.

### Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Metal fumes and oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

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## Product Information

### (a) acute toxicity;

Oral

Category 4

Dermal

Based on available data, the classification criteria are not met

Inhalation

Category 4

## Toxicology data for the components

| Component         | LD50 Oral                                  | LD50 Dermal                   | LC50 Inhalation       |
|-------------------|--------------------------------------------|-------------------------------|-----------------------|
| Toluene           | > 5000 mg/kg ( Rat )                       | LD50 = 12000 mg/kg ( Rabbit ) | 26700 ppm ( Rat ) 1 h |
| Isopropyl alcohol | 5045 mg/kg ( Rat )<br>3600 mg/kg ( Mouse ) | 12800 mg/kg ( Rat )           | 72.6 mg/L ( Rat ) 4 h |

### (b) skin corrosion/irritation;

Category 2

### (c) serious eye damage/irritation;

Category 2

### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

No information available

### (e) germ cell mutagenicity;

No data available

### (f) carcinogenicity;

No data available

There are no known carcinogenic chemicals in this product

### (g) reproductive toxicity;

Category 1A

### (h) STOT-single exposure;

No data available

Results / Target organs

Central nervous system (CNS).

### (i) STOT-repeated exposure;

Category 2

Target Organs

Central nervous system (CNS), Kidney, Blood.

### (j) aspiration hazard;

Category 1

### Symptoms / effects, both acute and delayed

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

### Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

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| Component         | Freshwater Fish                                                                                                                                                                                              | Water Flea                                                                                   | Freshwater Algae                                                                                                           | Microtox                                           |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Toluene           | 50-70 mg/L LC50 96 h<br>5-7 mg/L LC50 96 h<br>15-19 mg/L LC50 96 h<br>28 mg/L LC50 96 h<br>12 mg/L LC50 96 h                                                                                                 | EC50: = 11.5 mg/L, 48h (Daphnia magna)<br>EC50: 5.46 - 9.83 mg/L, 48h Static (Daphnia magna) | EC50: = 12.5 mg/L, 72h static (Pseudokirchneriella subcapitata)<br>EC50: > 433 mg/L, 96h (Pseudokirchneriella subcapitata) | EC50 = 19.7 mg/L 30 min                            |
| Isopropyl alcohol | LC50: = 9640 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: > 1400000 µg/L, 96h (Lepomis macrochirus)<br>LC50: = 11130 mg/L, 96h static (Pimephales promelas)<br>LC50: = 10000000 µg/L, 96h (Daphnia) | 13299 mg/L EC50 = 48 h<br>9714 mg/L EC50 = 24 h                                              | EC50: > 1000 mg/L, 72h (Desmodesmus subspicatus)<br>EC50: > 1000 mg/L, 96h (Desmodesmus subspicatus)                       | = 35390 mg/L EC50 Photobacterium phosphoreum 5 min |

**Persistence and degradability** No data is available on the product itself

| Component                  | Degradability |
|----------------------------|---------------|
| Toluene<br>108-88-3 ( 10 ) | 86% (20d)     |

**Degradation in sewage treatment plant**

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative potential** Product has a high potential to bioconcentrate

| Component         | log Pow | Bioconcentration factor (BCF) |
|-------------------|---------|-------------------------------|
| Toluene           | 2.73    | 90                            |
| Isopropyl alcohol | 0.05    | No data available             |

**Mobility in soil**

No information available.

**Endocrine Disruptor Information**

This product does not contain any known or suspected endocrine disruptors

**Other adverse effects**

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

**Waste treatment methods**

**Waste from Residues/Unused Products**

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations Should not be released into the environment

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous Keep product and empty container away from heat and sources of ignition

**Other Information**

Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Do not let this chemical enter the environment



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## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN1993  
Hazard Class 3  
Packing Group II  
Proper Shipping Name Flammable liquid, n.o.s. Isopropyl alcohol, Toluene

### Road and Rail Transport

UN-No UN1993  
Hazard Class 3  
Packing Group II  
Proper Shipping Name Flammable liquid, n.o.s. Isopropyl alcohol, Toluene

### IATA

UN-No UN1993  
Hazard Class 3  
Packing Group II  
Proper Shipping Name Flammable liquid, n.o.s. Isopropyl alcohol, Toluene

Special Precautions for User No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

| Component                                                | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|----------------------------------------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Lead titanium zirconium oxide (Pb(Ti,Zr)O <sub>3</sub> ) | 235-727-4 | X    | X   | -     | -    |      | -     | -    | KE-21950 |
| Toluene                                                  | 203-625-9 | X    | X   | X     | X    | X    | X     | X    | KE-33936 |
| Isopropyl alcohol                                        | 200-661-7 | X    | X   | X     | X    | X    | X     | X    | KE-29363 |

### Note

Note 1: The concentration stated or, in the absence of such concentrations, the generic concentrations of this Regulation (Table 3.1) or the generic concentrations of Directive 1999/45/EC (Table 3.2), are the percentages by weight of the metallic element calculated with reference to the total weight of the mixture

| Component                                                | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Lead titanium zirconium oxide (Pb(Ti,Zr)O <sub>3</sub> ) |                                                                                           |                                                                                          |                            | Annex I - Y31                      |
| Toluene                                                  |                                                                                           |                                                                                          |                            | Annex I - Y42                      |
| Isopropyl alcohol                                        |                                                                                           |                                                                                          |                            | Annex I - Y42                      |

### National Regulations

Persistent Organic Pollutant This product does not contain any known or suspected substance  
Ozone Depletion Potential This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

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## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Prepared By**

Health, Safety and Environmental Department

**Revision Date**

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**Revision Summary**

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**